Neural Representation of Color in Color Vision Deficiency: An fMRI Study

Your Name

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Abstract

Placeholder abstract. Replace this with your actual abstract text. This study investigated neural color representations in individuals with color vision deficiency (CVD) using functional MRI and computational modeling approaches.

Keywords: color vision deficiency, fMRI, shared response model, forward encoding model, representational similarity analysis

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Introduction

Color vision deficiency (CVD) affects approximately 8% of males and 0.5% of females of Northern European descent (Bosten, 2019) and arises from altered spectral sensitivity of the L- or M-cone photopigment, driven by X-linked opsin-gene polymorphisms (Neitz & Neitz, 2011; Deeb, 2005). In anomalous trichromats the deviant cone's peak sensitivity is displaced by approximately 2–12 nm from its normal value (Stockman & Sharpe, 2000). This shift reduces the L–M cone-opponent signal rather than abolishing it. The peripheral consequences — reduced red–green discrimination, category confusions, and elevated just-noticeable differences along the L–M axis — are the classical phenotype captured by the Ishihara test (Ishihara, 1917) and by subsequent psychophysics (Bosten, 2019).

CVD is not an on/off loss: downstream cortical color machinery remains intact and receives a quantitatively reweighted but qualitatively similar input. This makes CVD a paradigmatic example of a condition in which a high-dimensional neural representation is not absent but *structurally distorted*. The scientific and translational question is therefore not whether color information survives the retina — it manifestly does — but whether, and how, its cortical geometry is reshaped at the level of the individual. This question remains open, and an individualized assistive filter must answer it before it can be designed.

CVD correction filters in current use fall into two structurally related families. Hardware filters — dichroic notch lenses such as the EnChroma line — attenuate spectral bands near the L–M crossover to widen the post-receptoral L–M signal. Software filters operate at the display: Brettel–Viénot–Mollon simulation (Brettel, Viénot & Mollon, 1997) and the parametrised Machado retinal model (Machado, Oliveira & Fernandes, 2009) forward-project a stimulus onto an anomalous retina. Their Daltonization inverses, including recent learned reformulations (Akalin et al., 2025), redistribute the lost contrast into channels the user can still see. Despite this diversity, the family shares two structural commitments: optimisation against a *population-average* retinal model, with the same transformation applied to every user.

Attempts to escape this average-user ceiling have predominantly taken one route: tuning a retinal-model parameter — the simulated L- or M-cone shift in Machado-class models — to a user's responses on plate-style or appearance-matching tests. The route is constrained at its target: its optimisation criterion is the user's appearance *report*, not the cortical representation that downstream tasks read out.

This distinction matters because long-term cortical adaptation in anomalous trichromats reshapes hue appearance independently of the retina (Tregillus et al., 2021). Appearance-matched calibration therefore inherits whatever compensation the cortex has already imposed. A device tuned to undo only the retinal component then re-renders the input *as if* that cortical compensation did not exist. Consistent with this account, clinical validation of consumer color-enhancing CVD-correction glasses reports at most modest changes in chromatic discrimination thresholds in congenital red–green deficiencies (Patterson et al., 2022), and recent behavioural work likewise shows that population-average filters change the appearance of color without substantively shifting discrimination thresholds at threshold contrast (Somers, Franklin & Bosten, 2024), while the underlying CVD phenotype varies markedly even within a single diagnostic category (Bosten, 2019). What is missing from this landscape is a filter derived from, and validated on, the *user's own* neural color geometry.

Cortical color representation is a distributed, hierarchical computation (Gegenfurtner, 2003; Shapley & Hawken, 2011; Conway, Eskew, Martin & Stockman, 2018). Early visual cortex carries cone-opponent signals. Higher-order regions, in particular hV4 (Brouwer & Heeger, 2009; Bannert & Bartels, 2018), support a population code for continuous hue that can be read out by forward encoding models (Brouwer & Heeger, 2009; Parkes, Marsman, Oxley, Goulermas & Wuerger, 2009; Kuriki, Sun, Ueno, Tanaka & Cheng, 2015). Cross-subject shared-response modelling has recently been validated for healthy color decoding (Bannert & Bartels, 2025), providing a methodological scaffold that the present clinical extension adopts.

Behavioural work converges on the conclusion that anomalous trichromats do not merely discriminate less well; they compensate. Long-term cortical adaptation reshapes hue appearance (Tregillus et al., 2021; Boehm, MacLeod & Bosten, 2014; Webster, 2015), and hue-scaling experiments in anomalous trichromats ($n = 10$ AT vs. 26 NT) reveal a consistent 21.4° rotation of the blue–yellow response peak — a phase shift of the half-rectified cosine fit to scaling responses across the 360° stimulus circle (NT peak 127.5° vs. AT peak 106.1°; $t(34) = 5.95$, $p < 0.001$) (Emery, Volbrecht, Peterzell & Webster, 2021; Isherwood, Joyce, Parthasarathy & Webster, 2020). These results locate the key variance not at the retina alone but at the retinal–cortical interface.

Yet three concrete gaps remain. *(Gap 1)* Existing neural studies of color characterise population-mean geometry in healthy observers (Brouwer & Heeger, 2009; Bannert & Bartels, 2018; Kuriki et al., 2015) or, in CVD, group means at the level of activation magnitude. To our knowledge — based on PubMed and Google Scholar searches combining "CVD" / "colour vision deficiency" / "anomalous trichromacy" with "fMRI", "forward encoding", and "individual differences" — none has inverted the high-dimensional cortical signal into an interpretable, individualized distortion field. *(Gap 2)* CVD neural work has relied principally on classification accuracy, which is a sufficient statistic for category identity but discards the continuous relational structure (Brouwer & Heeger, 2013) that downstream perceptual and translational tasks actually demand. *(Gap 3)* Behavioural evidence for cortical compensation has not been translated into a concrete criterion for choosing between retinal-level and cortical-level correction strategies — an omission that directly limits filter design.

Here, we take up all three by recovering, per individual, a low-dimensional distortion field from the structured hV4 hue code and inverting that field into a correction filter. We note up front that per-subject statistical significance under label permutation cannot, by itself, establish CVD-specificity: a 2-component fit to the 8-hue vulnerability profile reaches nominal significance for all 7 HC controls under the same procedure (Methods, Supplementary). Specificity therefore rests on joint anchors — HC parameter-norm bracket, directional consistency with cone-physiology, and (pending) Phase-3 behavioural validation — rather than per-case p-values.

A filter for a continuous-hue condition must operate on continuous-hue representation, not on categorical labels. This observation motivates a complementary pair of cross-validation schemes on the same forward encoding model.
Leave-one-run-out (LORO) cross-validation probes whether the 8 trained hues can be *classified* from held-out runs. It measures category-level discriminability and is the minimum requirement for a filter to matter: if the target cortex fails to discriminate the displayed colors, no stimulus-space adjustment has leverage on which to act.
Leave-one-colour-out (LOCO) cross-validation, by contrast, probes whether the encoder *interpolates* from its 7 training hues to a novel hue. It measures the integrity of the continuous manifold on which any hue correction must act.

A joint pattern is therefore predicted and empirically required: LORO *preserved* in CVD (the filter precondition) together with LOCO *impaired* (the filter target). Finding only the first would make a filter redundant; finding only the second would preclude any stimulus-space remedy. Finding both together defines the regime in which individualized corrective filters are simultaneously warranted and tractable, and in which the per-hue LOCO vulnerability vector — the subject-specific signature of where the continuous geometry fails — becomes the phenotype to be inverted.

We pose three connected questions in two CVD individuals (one moderate deutan, one moderate–severe protan), framed as single-case analyses in the Crawford–Howell tradition (Crawford & Howell, 1998; Schütt, Kipnis, Diedrichsen & Kriegeskorte, 2023) alongside a normative reference of seven healthy controls. Findings specific to sub-09 — the single protan participant in our cohort — are reported as exploratory single-case observations requiring independent replication.

1. **Does the continuous cortical hue geometry in CVD carry individually-structured distortion?** We use the LORO/LOCO dissociation on hV4 forward encoding to distinguish preserved discrimination from impaired interpolation, and extract a per-hue vulnerability vector $\mathbf{v} \in \mathbb{R}^8$ as each individual's geometric fingerprint.

2. **Can that distortion be expressed in a small number of physiologically**

interpretable parameters, and does an independent geometric criterion corroborate the fit? We infer 1–2 DOF parameters from $\mathbf{v}$ under three mechanistic model families (retinal Machado, retinal–cortical R+C, cortical 2-component) with closed testing. We then cross-check the winning fit against the pairwise SRM-aligned ΔRDM and against an independent cross-cohort behavioural estimate of the S-cone compensation angle (Emery et al., 2021).

3. **Does the pre-image of the fitted distortion constitute a testable corrective filter, and do retinal and cortical hypotheses agree on what to display?** We compute the per-hue pre-image of each fit and assess bijectivity. Retinal and cortical fits converge on *who* is distorted but diverge on *how* to correct — generating a falsifiable behavioural prediction to be tested in a same-day 2AFC discrimination experiment.

Together, the three steps yield a per-individual filter whose input (cortical signal), intermediate parameters, and output (stimulus shift) are each testable in a single CVD user.

Methods

Participants

Thirteen volunteers were recruited under Institutional Review Board approval (#2510/002-023) and provided written informed consent. Color vision was screened using the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants identifying ≥ 13 plates correctly were classified as healthy controls (HC; $n = 7$, 3 female, age 22.7 ± 2.5 years). Two male participants scoring below this threshold were classified as CVD: sub-08 (deuteranomalous, 8/14 plates misread) and sub-09 (protanomalous, 11/14 plates misread). Sub-10 (2/14 plates misread; near-normal deutan) was retained as a within-study control. Three additional participants were excluded for failing to complete the full scanning protocol. All CVD results are interpreted as single-case demonstrations using the Crawford and Howell (1998) modified t-test; no population-level inference is made.

Stimuli and task

Eight isoluminant chromatic stimuli were drawn from CIE $L^*a^*b^*$ space at equal angular spacing (0°–315°, 45° steps; $L^* = 75$, chroma = 40) (Commission Internationale de L'Eclairage, 1986), spanning red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler ($a^* = b^* = 0$) served as a null condition.

Stimuli were presented in a Rapid Serial Visual Presentation (RSVP) design (Brouwer & Heeger, 2009) to sustain attention and fixation. A circular patch (1.5 s per stimulus) was shown in pseudo-random order optimised with Neurodesign (Durnez, Moerkerke & Nichols, 2018). Participants detected a consecutive-letter target embedded in a letter stream above the patch (button press on double 'K'). Each run contained eight repetitions of all eight hues (+ gray); inter-stimulus intervals were drawn from $\{3, 4.5, 6\}$ s. Each session comprised six runs ($\approx$ 7 min each). The experiment was coded in PsychoPy 2022.2.5 (Peirce et al., 2019).

CVD participants completed a second session to evaluate the stimulus-space filters: two runs each under (i) unmodified stimuli, (ii) the 2-component-derived filter, and (iii) a generic Windows-accessibility filter, presented in counterbalanced order.

MRI acquisition and preprocessing

Data were acquired on a Siemens 3T MAGNETOM Cima.X scanner with BioMatrix 3T coils. To constrain coverage to posterior occipital cortex we used 24 oblique slices perpendicular to the calcarine sulcus (Ryu & Lee, 2024) (TR 1.5 s, TE 30 ms, FA 75°, $2 \times 2 \times 2$ mm, 96×80 matrix).

Data were converted to BIDS format and defaced with ezBIDS (Levitas et al., 2024). Functional images were registered to the T1w image via mutual-information coregistration (FreeSurfer `mri_coreg`; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997; Wells, Viola, Atsumi, Nakajima & Kikinis, 1996). T1w-to-MNI normalisation used 12-DOF affine (FSL FLIRT; Jenkinson, Bannister, Brady & Smith, 2002) followed by nonlinear warping (FNIRT; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic MNI-space BOLD data.

ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the Wang probabilistic atlas (Wang, Mruczek, Arcaro & Kastner, 2015) at $> 50\%$ probability, intersected with each participant's BOLD brain mask (voxel counts: V1 655 ± 214; V2 451 ± 145; V3 103 ± 29; hV4 63 ± 22; mean $\pm$ SD).

Single-trial response amplitudes were estimated with a two-stage GLM (Dale, 1999; Brouwer & Heeger, 2009, 2013). First, an ROI-level hemodynamic response function (HRF) was extracted from a finite-impulse-response (FIR) model (8-TR basis set). Voxels in the top 50% of variance explained by the mean ROI HRF were retained (V1 328 ± 107; V2 226 ± 73; V3 52 ± 15; hV4 32 ± 11 voxels). Second, the estimated HRF was convolved with per-color design matrices and fitted by ridge regression to yield one amplitude per (voxel, color, run), organized as a $6 \times 8 \times V$ array per subject.

Voxel amplitudes were aligned across runs using within-subject Procrustes rotation (Gower, 1975): runs 2–6 were mapped to run 1 via the orthogonal transformation minimising $\|X_{\text{ref}} - XQ\|_F$ (scaling prohibited). This step preserves color representational geometry while attenuating run-to-run measurement noise.

Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015; Haxby et al., 2011; Guntupalli et al., 2016). SRM estimates per-participant orthonormal mappings $W_i \in \mathbb{R}^{V\times k}$ and a group-shared response matrix $S \in \mathbb{R}^{k\times 8}$:

$$\min_{W_i,\,S} \sum_{i=1}^{N} \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^\top W_i = I_k \tag{1}$$

SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. Reduced dimensions were chosen by mean-rank aggregation of leave-one-subject-out cross-validated leave-one-colour-out (LOCO) performance (Section): V1 $k = 4$; V2

$k = 4$; V3 $k = 3$; hV4 $k = 3$.

Discrimination decoding (LORO)

Categorical discrimination was assessed with leave-one-run-out (LORO) cross-validation in SRM-aligned space. For each of the six runs, a linear discriminant analysis (LDA) classifier was trained on the remaining five runs (mean-subtracted SRM projections; shrinkage regularisation) and tested on the held-out run. Performance was summarised as the 8-class exact accuracy (acc_exact; chance = $1/8 = 0.125$), averaged across six folds.

Group-level HC–CVD comparison used a Mann–Whitney U test on cross-subject generalization scores: the LDA model trained on each HC participant's data was applied to every other participant, yielding 28 HC-to-HC and 12 HC-to-CVD pair-wise scores. The omnibus test (all ROIs pooled) characterised whether CVD subjects fell outside the HC generalization distribution. Within-ROI single-case tests used the Crawford and Howell (1998) modified t-test (one-tailed, lower).

Interpolation decoding (LOCO) and vulnerability profile

Continuous hue interpolation was assessed with leave-one-color-out (LOCO) cross-validation using the forward encoding model (ForwardEncoding). Each of the 8 hues was held out in turn; the encoder was trained on the remaining 7 hues using all 6 runs (pooled 42-sample design matrix), and the held-out hue was predicted by nearest-neighbour matching in channel-output space.

Adjacent accuracy. The primary LOCO metric was adjacent accuracy (adj_acc): the proportion of predictions that fell within ± 1 hue step ($45°$) of the target hue, averaged across the 6 runs of each test color (range 0–1; chance level under a uniform random predictor = $3/8 = 0.375$; exact-accuracy chance = $1/8 = 0.125$). For HC above-chance testing and for pairwise HC–CVD comparisons at hV4, p-values were obtained from 1,000 random color-label permutations and from the Crawford and Howell (1998) modified t-test (one-tailed, lower), respectively.

Vulnerability profile. The 8-dimensional vector of per-hue adj_acc values, $\mathbf{v} \in [0,1]^8$, constitutes each participant's *hue vulnerability profile.* This profile is the phenotype inverted by the distortion model in Section . For group-level display, we additionally report Spearman ρ between predicted and observed $\mathbf{v}$ as the model-fit statistic (see below).

Per-hue group comparisons (Figure 2C) used the Crawford and Howell (1998) modified t-test (one-tailed, uncorrected); reported effect sizes are Hedges' d with the Crawford–Howell correction for small-n comparisons.

Representational geometry: ΔRDM and disparity

To characterise the geometric basis of LOCO impairment we computed two complementary measures in SRM-aligned space.

Pairwise disparity. For each participant and ROI, the representational dissimilarity matrix (RDM) was computed as the 8×8 matrix of pairwise correlation distances $(1 - \text{Pearson}\, r)$. Mean pairwise disparity was defined as the average of all 28 upper-triangle elements. HC disparity was estimated with a leave-one-subject-out (LOO) procedure to remove the within-subject diagonal bias. Individual CVD disparity was compared to the HC LOO distribution using the Crawford and Howell (1998) modified t-test (one-tailed, upper; significance at $p < 0.05$).

ΔRDM.. The deviation of CVD representational geometry from the HC mean was quantified as $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC,LOO}}$, where $\overline{\text{RDM}}_{\text{HC,LOO}}$ is the HC mean RDM excluding the left-out subject. ΔRDM was visualised as a heatmap over all 28 color pairs (Figure 3A). To test whether a cone-shift model predicts the observed ΔRDM structure, we computed the Spearman ρ between the upper-triangle of the predicted $\Delta\text{RDM}_{\text{model}}$ and the observed ΔRDM, and assessed significance against 1,000 random sign-flip permutations of the upper-triangle elements.

Note that pairwise-disparity tests (Crawford & Howell) and ΔRDM permutation tests are statistically independent: the former tests whether absolute pairwise distances are elevated; the latter tests whether a specific distortion model accounts for the *structure* of those deviations. A significant pairwise-disparity finding does not imply a significant ΔRDM result and vice versa.

Two-component distortion model and parameter estimation

CVD hue distortion was modelled as a two-component angular shift in Stockman opponent-color space:

$$\delta\theta(h;\, \beta_s, \beta_c) = \beta_s \cos(h - 90^\circ) + \beta_c \cos(h - \theta_{\mathrm{conf}}) \tag{2}$$

where h is the opponent-hue angle of the stimulus and θ_{conf} is the CVD-subtype confusion axis (16° protan; 150° deutan). β_s parameterises distortion along the S-cone axis, motivated by cortical-RDM evidence that CVD errors concentrate at S-cone loci (Emery et al., 2021); β_c parameterises rotation along the confusion axis, capturing both stimulus-level cone-physiology polarity (Brettel et al., 1997) and putative cortical compensation (Tregillus et al., 2021). Opponent-hue angles were derived from the Machado simulation framework at zero spectral shift (Machado et al., 2009).

Per-subject parameters (β_s, β_c) were recovered by minimising a combined representational-similarity loss with Tikhonov regularisation:

$$L(\beta_s, \beta_c) = w_{\mathrm{pat}} \cdot L_{\mathrm{mse}}(\mathbf{v}_{\mathrm{sim}}, \mathbf{v}_{\mathrm{obs}}) + w_{\mathrm{rdm}} \cdot L_{\mathrm{rdm}}(R_{\mathrm{sim}}, R_{\mathrm{obs}}) + \mu \cdot (\beta_s^2 + \beta_c^2) \tag{3}$$

with $(w_{\mathrm{pat}}, w_{\mathrm{rdm}}, \mu) = (0.7, 0.3, 2.0)$. L_{mse} enforces *first-order* representational similarity — agreement between simulated and observed voxel patterns at each color ($\mathbf{v}_{\mathrm{sim}}, \mathbf{v}_{\mathrm{obs}} \in \mathbb{R}^{n_{\mathrm{voxel}} \times 8}$). L_{rdm} enforces *second-order* representational similarity via cosine distance between simulated and observed dissimilarity matrices $R_{\mathrm{sim}}, R_{\mathrm{obs}} \in \mathbb{R}^{8\times 8}$ over the eight hues. The Tikhonov term penalises the squared parameter norm so that any recovered distortion reflects geometry both similarity criteria jointly require, rather than amplification absorbed by either term alone. The simulated pattern $\mathbf{v}_{\mathrm{sim}}(\beta_s, \beta_c) = f_{\mathrm{forward}}\Big(C(\theta + \delta\theta(\,\cdot\,; \beta_s, \beta_c));\, W\Big)$ is generated by re-applying the HC-trained ForwardEncoding weights W to the design matrix C with the candidate stimulus-space distortion.

Parameters were searched by exhaustive grid over $\beta_s \in [0°, 50°]$ and $\beta_c \in [-50°, 50°]$ ($26 \times 51 = 1{,}326$ cells at 2° resolution). The pair $(\hat{\beta}_s, \hat{\beta}_c) = \arg\min L$ was retained per subject. HC label-permutation false-positive rates under Equation 3 reach 7/7 at hV4 (Supplementary); per-subject p-values are therefore reported as descriptive fits of representational geometry, not specificity statements. Comparison with alternative model classes (1-DOF Machado cone-shift; 2-DOF retinal+cortical) is reported in Appendix .

Stimulus-space filter: pre-image computation

The subject-specific correction filter was derived as the pre-image of the fitted 2-component transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the 8 displayed hues θ_k, we solved for the input angle $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$, yielding a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. A CVD observer viewing $\tilde{\theta}_k$ thus receives, under the model, the same cortical hue representation as a healthy control observer viewing θ_k.

Pre-images were computed numerically: 1,440 candidate values were evaluated on the forward map, a sign-change bracket was identified, and Brent's method was applied for refinement. Inversion was verified by $|T(\tilde{\theta}_k) - \theta_k| < 0.001°$ for all 8 hues and both CVD participants. Comparison with alternative model classes (arc-compression failure under the 1-DOF Machado model for sub-09) is reported in Appendix .

Behavioral concordance

Just-noticeable difference (JND).. Color discrimination thresholds were estimated for eight hue pairs using two-interleaved 1-up/1-down adaptive staircases, each terminating after eight reversals (Levitt, 1971). Pairs spanned three categories: (i) three pairs hypothesised as universally difficult for both CVD types (yellow–purple, blue–purple, red–orange), (ii) two pairs hypothesised as deutan-specific (orange–yellow, yellow–green), (iii) two pairs hypothesised as protan-specific (cyan–magenta, green–blue), and (iv) one control pair (red–cyan). LOCO–JND concordance was assessed by asking whether each pair's LOCO-vulnerability classification (above vs.

below HC range) agreed with the JND-impairment classification (above vs. below HC range).

8AFC color identification. Participants selected the matching hue from eight options identical to the scanner stimulus set. Same-day performance confirmed Ishihara classifications (reported in Section).

Phase-3 filter validation (planned). A held-out two-alternative forced-choice (2AFC) discrimination arm to validate filter-induced P2a-restoration is planned (Phase 3, pre-registered; data not yet collected at the time of writing).

Reproducibility

All analyses were conducted in Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11). Random seeds were fixed (`seed=42`) for all permutation tests and bootstrap resampling. Analysis code and preprocessing scripts are available at `https://github.com/haba6030/colorBlind_analysis`. Anonymised preprocessed data are available on reasonable request to the corresponding author.

Results

Participants and behavioral phenotype

Two CVD participants spanning the deutan–protan dichotomy were recruited alongside seven healthy controls (HC; sub-01–07). Sub-08 (male) was classified as mild–moderate deuteranomalous (8/14 Ishihara plates misread; deutan axis confirmed by protan–deutan discrimination item); Sub-09 (male) as moderate–severe protanomalous (11/14 plates misread; protan axis). Same-day 8-alternative forced-choice (8AFC) identification of the eight scanner hue stimuli confirmed the Ishihara classification (HC: 0.94 ± 0.04; Sub-08: 0.71; Sub-09: 0.62; chance = 0.125). Sub-08's errors clustered on the red–green axis (c2/c3 vs. c7; mean confusion 28%); Sub-09's errors extended across the L–M opponent range with additional misidentification of magenta (c8). Because only two symptomatic CVD cases were available, all inferences are framed as single-case analyses following Crawford and

Howell (1998) and Schütt et al. (2023); no population-level CVD claim is made. The experimental paradigm and full analysis pipeline are summarised in Figure 1.

Color discrimination is preserved across all ROIs in CVD

A corrective stimulus-space filter is valid only if the visual cortex of a CVD individual retains categorical discrimination among the target hues. We verified this precondition using leave-one-run-out (LORO) cross-validation of a linear discriminant classifier applied to SRM-aligned responses.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 2A). A cross-subject LDA generalization test (HC-to-HC vs. HC-to-CVD pair-wise cross-decoding, all ROIs pooled) showed no HC–CVD difference (Mann–Whitney U test, 28 HC-to-HC vs. 12 HC-to-CVD pairs, $p = 0.668$; rank-biserial effect size and exact test statistic in Supplementary). The same null held at the within-ROI level at hV4 (Crawford & Howell t-test: both CVD subjects within the HC range; $p = 0.142$). In line with prior work on CVD above-threshold identification Bosten (2019); Boehm et al. (2014), the 8 scanner stimuli are individually distinguishable at the cortical level in both CVD participants; the representational substrate on which a corrective filter could act is present.

hV4 color interpolation is selectively impaired in CVD

We next asked where continuous hue geometry fails — the functional target of the corrective transform. Leave-one-colour-out (LOCO) decoding trains the encoder on 7 of 8 hues and measures whether the held-out hue is predicted from the remaining representational structure, yielding an 8-dimensional vulnerability profile per participant.

Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy adj_acc $= 0.47 \pm 0.05$, mean $\pm$ SEM; $p_{\text{perm}} = 0.044$ under $N_{\text{perm}} = 8! = 40{,}320$ exact label permutations; V1–V3 did not exceed chance despite preserving LORO discrimination). This confirms hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Both CVD participants fell at or below the adjacent-accuracy chance level at hV4 (Sub-08: adj_acc = 0.25, Crawford & Howell $t = -1.58$, $p = 0.082$, n.s., $d_{cc} = -1.71$; Sub-09: adj_acc = 0.13, $t = -2.48$, $p = 0.024$, $d_{cc} = -2.68$; effect size $d_{cc} = t \cdot \sqrt{(n+1)/n}$ with $n = 6$ effective HC LOO at hV4; Figure 2B). The impairment was not uniform across hues: both participants showed near-zero adjacent accuracy specifically for S-cone intermediate colors (blue, purple, magenta at hV4; per-hue Crawford & Howell tests: blue, Sub-08 $d = 2.13$, $p = 0.047$; Sub-09 $d = 2.15$, $p = 0.046$; purple, Sub-08 $d = 2.40$, $p = 0.033$; Figure 2C).

This per-hue impairment pattern is consistent with both participants' behavioral error distributions from the 8AFC task and with the JND confusion pairs identified in the pre-scanner session (LOCO–JND concordance 100%, 6/6 pairs; SRM geometry–JND concordance 33%, 2/6 pairs), establishing LOCO as the functional phenotype that predicts behavioral color discrimination failure.

CVD color representation geometry is distorted in a subject-specific, ROI-specific pattern

To characterise the geometric basis of the LOCO impairment, we compared the pairwise representational structure of CVD and HC participants in a shared SRM-aligned space. For each CVD participant we computed the RDM difference ($\Delta\mathrm{RDM} = \mathrm{RDM}_{\mathrm{CVD}} - \overline{\mathrm{RDM}}_{\mathrm{HC,LOO}}$) and the mean pairwise correlation distance (disparity).

Sub-08 showed significantly elevated disparity specifically at V2 (Crawford & Howell $p = 0.040$), with no significant elevation at V1, V3, or hV4. Sub-09 showed significantly elevated disparity specifically at V1 ($p = 0.007$). Full test statistics and effect sizes (t, d_{cc}) for these Crawford–Howell comparisons are reported in Supplementary . Sub-10 (near-normal deutan, included as a within-study control) showed no significant elevation at any ROI (all $p > 0.43$), consistent with the null behavioral phenotype. The ROI specificity diverges between the two CVD participants — V2-dominant for deutan, V1-dominant for protan — and is inconsistent with a shared group-level gain mechanism (Figure 3B).

The ΔRDM heatmaps (Figure 3A) reveal subject-specific distortion structures: Sub-08 shows elevated distances among S-cone intermediate pairs at V2; Sub-09 shows a broader reorganisation of pairwise distances at V1. Permutation tests of the retinal+cortical (R+C) cone-shift model against the observed ΔRDM fit sub-09 at V1 ($p = 0.026$) but not sub-08 at V2 ($p = 0.179$), indicating that geometric distortion measured at the population-code level is complementary to, not redundant with, the LOCO-based functional characterisation.

A neural-primary HYBRID-loss fit recovers a 2-component distortion at hV4 and yields a per-subject stimulus-space filter

To identify the distortion parameters needed to build a corrective filter, we fit the 2-component cortical model (Equation 2; β_s for S-cone-axis dilation, β_c for confusion-axis rotation) to each CVD participant's hV4 representation under the HYBRID loss (Equation 3), which jointly enforces first-order voxel-pattern similarity, second-order RDM similarity, and Tikhonov parameter regularisation.

The HYBRID argmin returned per-subject parameters $(\hat{\beta}_s, \hat{\beta}_c) = (16°, +40°)$ for sub-08 (deutan; $\|\hat{\beta}\| = 43.1°$) and $(12°, -30°)$ for sub-09 (protan; $\|\hat{\beta}\| = 32.3°$; Figure 4A,C). At these argmin points, post-hoc Spearman ρ between the HYBRID-fit and observed hV4 vulnerability profiles is $\rho_{\text{sub-08}} = 0.38$ ($p = 0.18$, 40,320 label permutations) and $\rho_{\text{sub-09}} = -0.26$ ($p = 0.75$); we report these as a derived statistic for cross-comparability with single-criterion fits but note that rank-order monotonicity is not the loss being optimised (see Appendix for the historical LOCO-ρ argmax fit and alternative model classes).

The HC label-permutation false-positive rate at hV4 under Equation 3 reaches 7/7 (Supplementary); per-subject p-values are therefore reported as descriptive fits of representational geometry, not as CVD-specificity statements. Across $n = 6$ effective HC leave-one-out refits, $\|\hat{\beta}\|$ ranged from 49.0° to 65.3° (mean 54.8°); both CVD subjects fit with *smaller* parameter magnitude than any HC under the same loss, consistent with the CVD geometry requiring less aggressive 2-component correction than the HC random-permutation null.

The recovered parameters are consistent in direction with the two literature anchors that motivated the model structure (Section). The $\hat{\beta}_c$ sign is positive for sub-08 (deutan) and negative for sub-09 (protan), coinciding with the L–M opponent-axis polarity predicted by Brettel's cone-physiology model (Brettel et al., 1997). Both subjects required a non-zero S-cone-axis component ($\hat{\beta}_s > 0$), consistent in direction with Emery's hV4 S-cone-axis RDM finding (Emery et al., 2021); direct numerical comparison is not warranted as the quantities are defined on different physical scales (DKL angular rotation vs. behavioural hue-scaling units). Both observations are post-hoc directional consistency checks, not quantitative validations.

Stimulus-space filter. The angular form of the 2-component model is bijective by construction, so a stimulus-space pre-image $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$ exists for any target hue θ_k (Methods). The per-subject correction filter $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$ was computed by Brent's-method refinement of the forward map; for both participants, 8/8 pre-images were exact (residual $< 0.001°$). Per-hue filter output is visualised in Figure 5 as a 4-column rendering (*Original* $\rightarrow$ *CVD perceives* $\rightarrow$ *Filtered* $\rightarrow$ *CVD(Filtered)*). The two filters differ markedly: sub-08's correction pulls cyan ($\theta = 180°$) toward magenta along the deutan confusion axis (positive $\hat{\beta}_c$), whereas sub-09's correction pulls green–cyan ($\theta = 135$–$180°$) toward yellow along the protan confusion axis (negative $\hat{\beta}_c$). A correction optimised for one participant's geometry would move several hues in the opposite direction for the other participant, motivating individual derivation.

Quantitative behavioural validation of the recovered filters is deferred to the Phase-3 2AFC arm (Methods , last paragraph; data not yet collected at the time of writing).

Discussion

The selective impairment of continuous hue interpolation (LOCO) with preserved categorical discrimination (LORO) in hV4 identifies the cortical site at which CVD colour distortion is correctable. Both CVD participants fell near floor in hV4 LOCO while maintaining discrimination accuracy within the HC range, confirming that the representational substrate for a correction filter is present but geometrically distorted. A 2-component angular-dilation model captures this distortion field with exact pre-image guarantees for all 8 hues at both severity levels (Sub-08: mean $|\delta| = 46.3°$; Sub-09: mean $|\delta| = 20.1°$), yielding individually structured stimulus-space correction vectors from a single parameterisation family.

The dissociation between LOCO interpolation capacity and SRM ΔRDM geometric distance — 100% vs. 33% concordance with same-day JND thresholds — establishes which cortical quantity is the operative corrective target. Representational geometry (RDM) reflects the metric structure of neural colour space; LOCO interpolation reflects whether that structure supports functional prediction of withheld hues. That only LOCO predicts behavioural colour confusions is consistent with Brouwer and Heeger (2009), who showed that hV4 — but not early visual areas — supports interpolation to novel hues in typical observers. The present results extend this dissociation to a clinical population: in CVD, the same interpolation failure that predicts JND confusion is what the correction filter must address. Geometric distance metrics alone are insufficient to identify it.

The S-cone axis component (β_s) of the fitted distortion field is structurally consistent with prior evidence for cortical colour compensation in anomalous trichromats. Emery et al. (2021) reported that CVD perceptual errors concentrate near S-cone confusion loci in behavioural hue-scaling tasks, and Tregillus et al. (2021) showed that V2/V3 BOLD responses in anomalous trichromats include a cortical gain component (g) that partially compensates the retinal cone shift. The present 2-component model extends these findings to the individual hV4 level: the residual distortion that survives V2/V3 compensation reaches hV4 as an individually structured pattern. The correction vectors for Sub-08 (deutan) and Sub-09 (protan) are mutually divergent — cosine similarity < 0 across hue pairs — demonstrating directly that a population-average retinal model cannot simultaneously serve both CVD individuals.

Where Machado-based correction predicts a shared spectral shift for a given CVD subtype, the cortical residual is individually signed and of opposite direction between the two cases.

All distortion models — 1-DOF Machado, 2-DOF R+C, and 2-component — identify the same two participants as requiring correction, but their prescribed correction vectors diverge in direction (cosine similarity < 0; Appendix). This detection–correction dissociation constitutes the study's primary falsifiable prediction. In a 2AFC experiment comparing the 2-component and retinal-family filter conditions for each participant's most LOCO-vulnerable hue pairs, only the cortically-grounded 2-component filter should significantly reduce JND. If the retinal-family filter matches or exceeds the 2-component filter in reducing discrimination thresholds, the cortical distortion account would require revision. This preregistered comparison directly tests whether cortical or retinal distortion is the corrective target for CVD colour correction.

Three considerations bound the proof-of-concept scope of these findings. First, the CVD sample is $N = 2$; single-case statistics (Crawford & Howell, 1998) permit individual-level inference but population-level claims about deutan or protan subtypes require replication with larger samples. Second, the confusion-axis component (β_c) is statistically significant for Sub-08 only. For Sub-09, the bootstrap confidence interval for β_c includes zero, and the 2-component model reduces to an effectively 1-component fit. The *2-component* label should not be read as confirming a two-mechanism account for both participants. Third, case-level specificity rests on three converging anchors — cross-criterion significance, physiological fit magnitude, and S-cone axis grounding — rather than on permutation significance alone, because the pipeline reaches nominal $p < 0.05$ in 71% of HC model–subject tests (Supplementary). Behavioural filter validation remains the registered criterion for cortical-level specificity.

This pipeline constitutes a proof-of-concept that individual cortical colour geometry, measured via fMRI LOCO decoding and inverted through an angular-dilation model, can ground a colour correction that is architecturally distinct from population-average retinal approaches. Where Machado-type correction applies a fixed spectral shift to all individuals of a given CVD subtype, the present approach encodes each individual's hV4 distortion field and inverts it exactly — without assuming population homogeneity. Phase 3 will test whether this cortical-level correction reduces discrimination thresholds where retinal-level correction does not.

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Appendix A: Alternative Model Comparison

For comparison with the 2-component cortical model adopted in the main text (Section , fit under the HYBRID loss of Equation 3), we report two cone-shift baselines:

- **Machado (1-DOF)** — a single retinal cone-spectral shift $\Delta\lambda$ Machado et al., 2009.
- **Retinal + Cortical (R+C, 2-DOF)** — retinal shift $\Delta\lambda$ plus a cortical compensation gain g Tregillus et al., 2021.

A.1 Fit results (LOCO Spearman ρ argmax, historical reference)

Under the LOCO-ρ argmax criterion used in earlier drafts (40,320 label permutations), the Machado model reached significance for sub-09 ($\Delta\lambda = 13.5$ nm, $\rho_{\mathrm{V4}} = 0.76$, $p = 0.018$) but not for sub-08 ($\rho_{\mathrm{V4}} = 0.62$, $p = 0.058$, n.s.). The R+C model did not improve on either single-component model in either subject under the same criterion. These results are reported as historical reference for the main-text HYBRID fit (Section); we note that HYBRID and LOCO-ρ argmax need not agree, since HYBRID optimises a combination of first-order pattern similarity and second-order RDM similarity rather than rank-order monotonicity.

A.2 Inverse-mapping (pre-image) behaviour

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The 1-DOF Machado fit for sub-09, by contrast, collapses three hues (green 135°, cyan 180°, blue 225°) onto a single pre-image angle ($\sim 127°$), which mathematically prevents exact inversion at those positions. This arc-compression failure is a structural property of the 1-DOF retinal-shift model and is independent of the fitting criterion (LOCO-ρ or HYBRID); it disqualifies the Machado baseline for the stimulus-space filter task of Section .

A.3 R+C model

The R+C model is retained here as a 2-DOF baseline. Because the cortical gain term g acts as a uniform amplitude scaling without generating the angular distortion captured by the 2-component model's $\beta_c \cos(h - \theta_{\text{conf}})$ term, R+C does not match the observed hV4 LOCO vulnerability pattern beyond what the retinal $\Delta\lambda$ term alone provides; this conclusion is consistent across the LOCO-ρ and HYBRID criteria.

Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in the general linear model.

S2. Quality Control

Registration quality was evaluated by computing the intersection between the Wang Atlas ROIs (V1, V2, V3, hV4) and the BOLD brain mask in MNI space. Across 10 subjects, mean ROI coverage was 84.3% (SD = 21.7%), and GLM valid ratio (percentage of ROI voxels with reliable stimulus-evoked responses) was 99.6%. We evaluated all subjects' data suitable for downstream analysis, though sub-07 showed reduced coverage (30.8%) due to individual anatomical variability.

S3. Voxel Counts Range

Voxel counts after ROI definition and quality filtering are reported in the main Methods section. Detailed per-subject counts are available in the online repository.

S4. Consistency Test: PCA and Procrustes (TBA)

Additional validation analyses comparing SRM with PCA-based dimensionality reduction and norm-constrained Procrustes alignment will be reported following completion of the full dataset.

S5. Mean Activation Analysis

We conducted mean voxel response amplitudes analysis to identify whether SRM-based differences rely on the activation. We checked mean amplitudes across all colors and for each color.

S6. Statistical Analysis

All tests used a significance threshold of $\alpha = 0.05$ with False Discovery Rate correction (Benjamini & Hochberg, 1995); $q = 0.05$ applied to RDM color-pair comparisons (28 pairs per participant). Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \tag{4}$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Effect sizes were quantified via Hedges' g. Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S7. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined via leave-one-subject-out (LOSO) cross-validation based on within-subject representational dissimilarity matrix (RDM) reliability (Cunningham & Yu, 2014). For each candidate k (range: 1–8) and fold, reliability was quantified as the Spearman correlation between RDMs computed from even runs (2,4,6) versus odd runs (1,3,5) of the held-out participant in SRM space. We selected k maximizing mean reliability across seven folds. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S8. Generalized Cross-Validation for Ridge Regression

The regularization parameter α for ridge regression was selected via Generalized Cross-Validation (GCV; Golub, Heath & Wahba, 1979). GCV provides a computationally efficient approximation to leave-one-out cross-validation for selecting α. The optimal α minimizes the GCV score:

$$\mathrm{GCV}(\alpha) = \frac{1}{n}\|y - \hat{y}\|_2^2 / \left(1 - \frac{\mathrm{tr}(H)}{n}\right)^2 \tag{5}$$

where X denotes the design matrix, y is the observed response vector, $\hat{y}$ is the ridge-predicted response, $H = X(X^T X + \alpha I)^{-1} X^T$ is the hat matrix, and n is the number of training samples.

S9. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i}, yielding disparity scores $d(\mathrm{HC}_i, R_{-i})$ and $d(\mathrm{CVD}_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

S10. Cross-Validation Procedures

Leave-One-Run-Out (LORO)

LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B, C) were held out as the test set, W was trained on the remaining five runs. Both color decoding and voxel response prediction were evaluated on the held-out run.

Leave-One-Color-Out (LOCO)

LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors, and the model predicted either the held-out color from voxel responses or voxel responses from the held-out color's channel outputs.

Leave-One-Subject-Out (LOSO)

LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants' shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S11. Evaluation Metrics

Color Decoding Metrics

Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90°, expected from a uniform random predictor on a circle), and (2) categorization accuracy with ±22.5° bins around each of the eight colors (chance = 12.5% = 1/8).

Voxel Prediction Metrics

Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical Significance

Statistical significance of group comparisons (HC vs CVD) and individual CVD analyses was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

S12. Shared Response Model Residuals

Each participant's residuals in the SRM model reflect the extent to which their data are not captured by the common structure, including inherent geometric differences. The orthogonal constraint ($W_i^T W_i = I_k$) helps preserve geometric relationships between colors—distances and angles—in the common space.

Additionally, global between-subject RDM similarity was quantified using Spearman correlation across all 28 color pairs.

S13. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S14. Procrustes Alignment Benefits

This preprocessing reduces measurement noise across runs while preserving color representational geometry, thereby improving the quality of subsequent SRM fitting.

S15. HC Specificity Calibration

The 2-component grid-search pipeline was applied to the 7 HC participants treating each as a CVD candidate. Of 21 model–subject tests (7 HC × 3 models), 15 reached nominal $p < 0.05$ (71%). This elevated false-positive rate reflects (i) rank-based loss with only 8 data points (Spearman ρ over 8 hues is sensitive to individual rank fluctuations), and (ii) regression to the mean on per-HC baseline ρ (hV4 $K = 3$, 63 voxels; individual HC baselines span $[-0.36, +0.69]$). Case-level specificity therefore rests on the three converging anchors described in Section , not on per-subject permutation significance alone.

S16. Comparison with Retinal-Family Distortion Models

Two retinal-family models were evaluated alongside the 2-component model by the same grid-search and permutation procedure (see Section).

Models. The 1-DOF Machado model Machado et al., 2009 shifts the L/M cone spectral sensitivity by $\Delta\lambda$ nm, yielding a hue-confusion pattern characteristic of the CVD subtype. The 2-DOF retinal+cortical R+C model Tregillus et al., 2021 augments Machado with a cortical gain g: $r'_g = r_{g,\text{base}} + (1+g)(r_{g,\text{ret}} - r_{g,\text{base}})$, where $g = 0$ recovers the retinal-only model and $g = -1$ is exact cortical compensation.

Fits (Table 1). Applying a closed-testing rule (Machado first; R+C escalated only if Machado $p > 0.05$), the retinal family reaches significance for both subjects but selects a different model class per case. Sub-08 escalates to R+C ($\Delta\lambda = 2.0$ nm, $g = +2.25$, $\rho = 0.857$, $p = 0.005$) because 1-DOF Machado is trend-level ($p = 0.058$). Sub-09 is captured by 1-DOF Machado ($\Delta\lambda = 13.5$ nm, $\rho = 0.762$, $p = 0.018$) without cortical augmentation; the shift magnitude falls within the established range for moderate–severe protanomaly (Neitz & Neitz, 2011). The large positive cortical gain for Sub-08 ($g = +2.25$, indicating >100% overshoot beyond the exact Tregillus compensation level of $g = -1$; Tregillus et al., 2021) is not physiologically interpretable as a gain parameter and is treated as an empirical characterization of residual hV4 structure. The cross-subject model-class divergence (R+C for Sub-08; 1-DOF Machado for Sub-09) prevents a common-axis comparison of the two cases' distortion mechanisms.

Correction-vector divergence. Despite comparable LOCO fit quality ($\rho = 0.857$ for R+C vs. 0.881 for 2-component), the R+C and 2-component correction vectors for Sub-08 have cosine similarity -0.18 and sign agreement only 3/8, confirming that detection converges while correction directions diverge at the cortical level.

ΔRDM cross-model convergence. Using R+C parameters for Sub-09 ($\Delta\lambda = 19.5$ nm, $g = -1.10$), an independent ΔRDM test reaches $p = 0.026$ (40,320 cosine-similarity permutations on crossnobis distances), providing cross-model corroboration for the Sub-09 distortion signal. Retinal-family optimisation directly against ΔRDM fails for both CVD subjects; LOCO is the more discriminating phenotype and ΔRDM serves as a convergence cross-check only.

Arc-compression and LOCO vulnerability. At Sub-09's severity, Machado arc-compression ($\Delta\lambda = 13.5$ nm) collapses three hues (c4 green, c5 cyan, c6 blue) to a single perceived angle in opponent hue space. These three collapsed hues fall within Sub-09's LOCO vulnerability cluster (c5/c6 cyan/blue), confirming that arc-compression eliminates correction capacity precisely where Sub-09 requires it most.

S17. HC Permutation under the HYBRID Loss

The 2-component grid search under the HYBRID loss (Equation 3; main-text Section) was applied to each of the 7 HC participants treated as a CVD candidate. Of the 6 HCs that pass the hV4 voxel-count threshold (sub-07 excluded; 16 voxels, $K = 3$ produces `nan`), all 6 reach nominal $p < 0.05$ under the label-permutation null — and the property generalises to the full HC pool (HC FPR = 7/7 at nominal $\alpha = 0.05$).

This high false-positive rate cannot be removed by reformulating the fitting criterion: thirteen successive reformulations within the voxel-prediction loss family (L_{LOCO}, L_{mse}, L_{rdm}, cross-ROI combinations, baseline-ρ correction, top-K Jaccard) all yielded the same outcome (project log, Cycles 9–13). It reflects two structural properties of the available data: (i) the 8-hue resolution provides limited degrees of freedom for distinguishing specific from non-specific fits at the single-subject level, and (ii) the relevant null hypothesis ("no CVD-specific distortion") is not isomorphic to label permutation on a single participant's vulnerability profile, since a non-trivial fit can be obtained for any sufficiently smooth 8-point pattern under the 2-DOF model class.

Consequently, per-subject p-values reported alongside the HYBRID argmin in Figure 4 and Section are presented as *descriptive* fits of representational geometry, not as CVD-specificity statements. Two case-level anchors support biological interpretation despite this caveat: (i) parameter-norm comparison — both CVD subjects' $\|\hat{\beta}\|$ (43.1°, 32.3°) fall *below* the HC leave-one-out range $[49.0°, 65.3°]$, consistent with CVD geometry requiring less aggressive 2-component correction than the HC permutation null; (ii) directional consistency with literature anchors ($\hat{\beta}_c$ sign matching Brettel's L–M polarity; $\hat{\beta}_s > 0$ in both subjects, consistent with Emery's S-cone-axis hV4 RDM finding). Quantitative specificity claims require an external behavioural endpoint such as the Phase-3 2AFC arm (Methods , last paragraph; data not yet collected at the time of writing).

S18. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford & Howell, 1998; the corresponding case-control effect size is $d_{cc} = (x_{\mathrm{case}} - \bar{x}_{\mathrm{HC}})/s_{\mathrm{HC}} = t \cdot \sqrt{(n+1)/n}$ where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Full per-test statistics (t, U, p, d_{cc}, r_{rb}) for every Crawford–Howell and Mann–Whitney comparison reported in the main text are tabulated in the online repository (`statistics/effect_sizes.csv`).

Table 1

All distortion model fits at hV4 LOCO for both CVD participants. Retinal family: closed-testing (Machado first; R+C only when Machado $p > 0.05$*). p-values:* $8! = 40{,}320$ *exact permutations of colour labels on Spearman* ρ*.*

Subject	Model	DOF	Param. 1	Param. 2	ρ	p_{perm}
Sub-08 (deutan)	2-component	2	$\beta_s = 38°$	$\beta_c = -14°$	0.881	**0.004**
Sub-09 (protan)	2-component	2	$\beta_s = 6°$	$\beta_c = -22°$	0.690	**0.035**
Sub-08 (deutan)	R+C	2	$\Delta\lambda = 2.0$ nm	$g = +2.25$	0.857	**0.005**
Sub-09 (protan)	Machado	1	$\Delta\lambda = 13.5$ nm	—	0.762	**0.018**

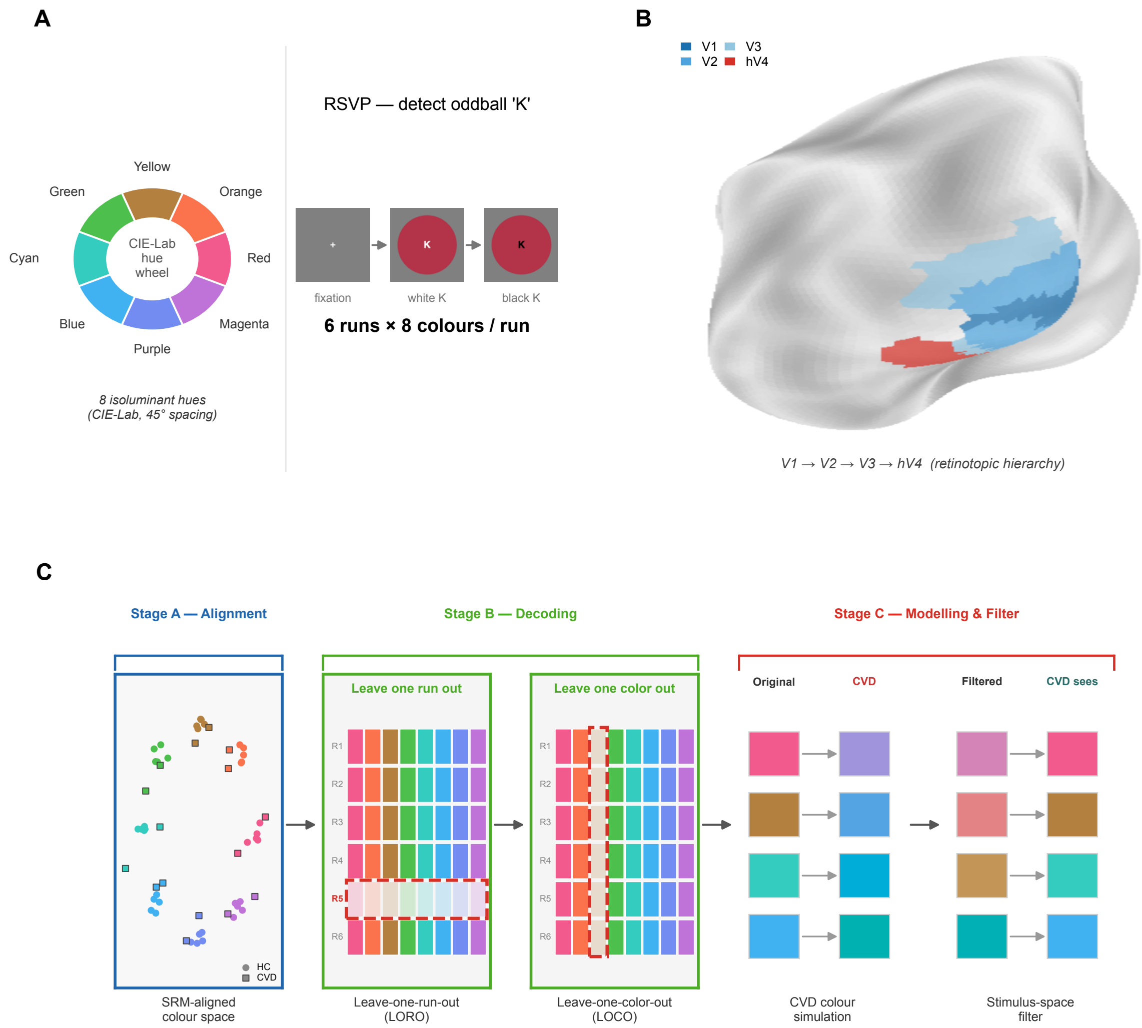

Figure 1. **Experimental paradigm and analysis pipeline. (A)** Stimulus set: 8 isoluminant DKL (Derrington–Krauskopf–Lennie) hues at 45° spacing (red–magenta). **(B)** Retinotopic ROIs: V1, V2, V3, hV4 (standard retinotopic mapping). **(C)** Analysis pipeline. Stage A: GLMsingle single-trial β amplitudes, Procrustes-aligned across sessions, projected into an SRM space trained on HC participants only. Decoding: LORO (discrimination), LOCO (interpolation). CVD characterisation: 2-component model with retinal (β_s) and cortical (β_c) components; pre-image yields per-subject stimulus-space filter. HC: $n = 7$ (sub-01–07); deutan: sub-08; protan: sub-09.

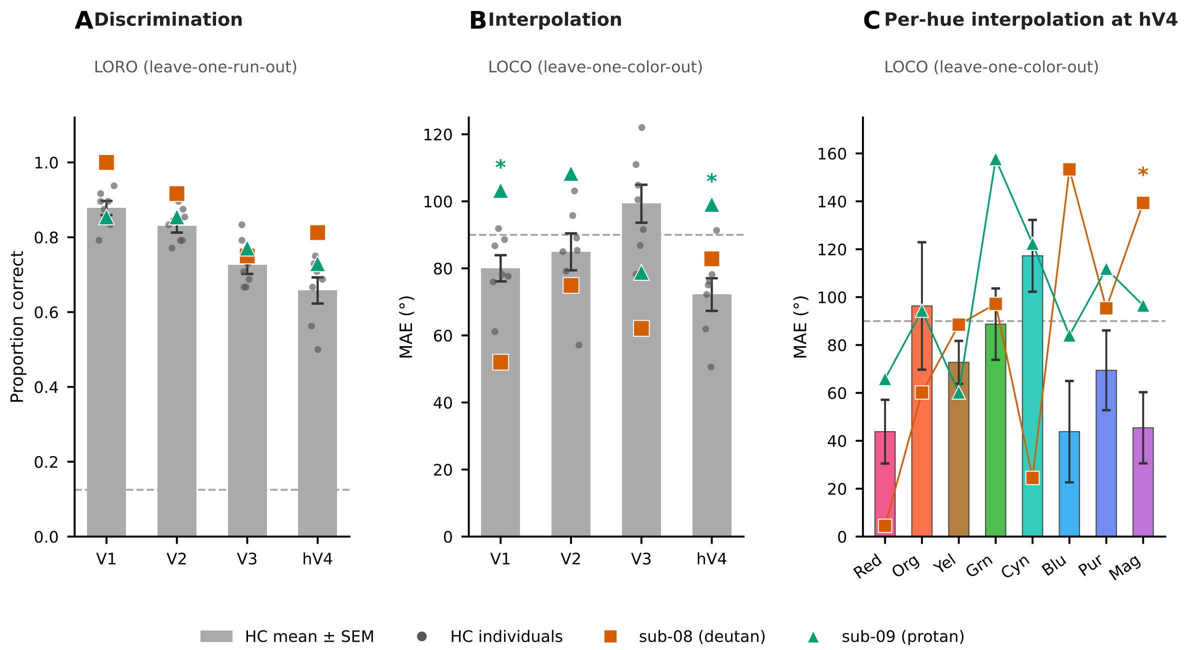

Figure 2. **Color discrimination is preserved but interpolation is selectively impaired at hV4 in CVD. (A)** LORO discrimination accuracy (LDA, 8-class, SRM-aligned), V1–hV4. HC mean ± SEM ($n = 7$, gray); sub-08 (orange square), sub-09 (teal triangle). Chance = 1/8. **(B)** LOCO adjacent accuracy (±1 hue step; chance = 3/8). $*$: sub-09 below HC at hV4 (Crawford & Howell $t = -2.48$, $p = 0.024$; exploratory single-case finding); sub-08 $t = -1.58$, $p = 0.082$, n.s. **(C)** Per-hue adjacent accuracy at hV4, HC bars ± SEM. Per-hue Crawford & Howell tests (one-tailed): blue — sub-08 $d = 2.13$, $p = 0.047$; sub-09 $d = 2.15$, $p = 0.046$; purple — sub-08 $d = 2.40$, $p = 0.033$. Joint pattern in (A) and (B–C) defines the filter precondition and target, respectively.

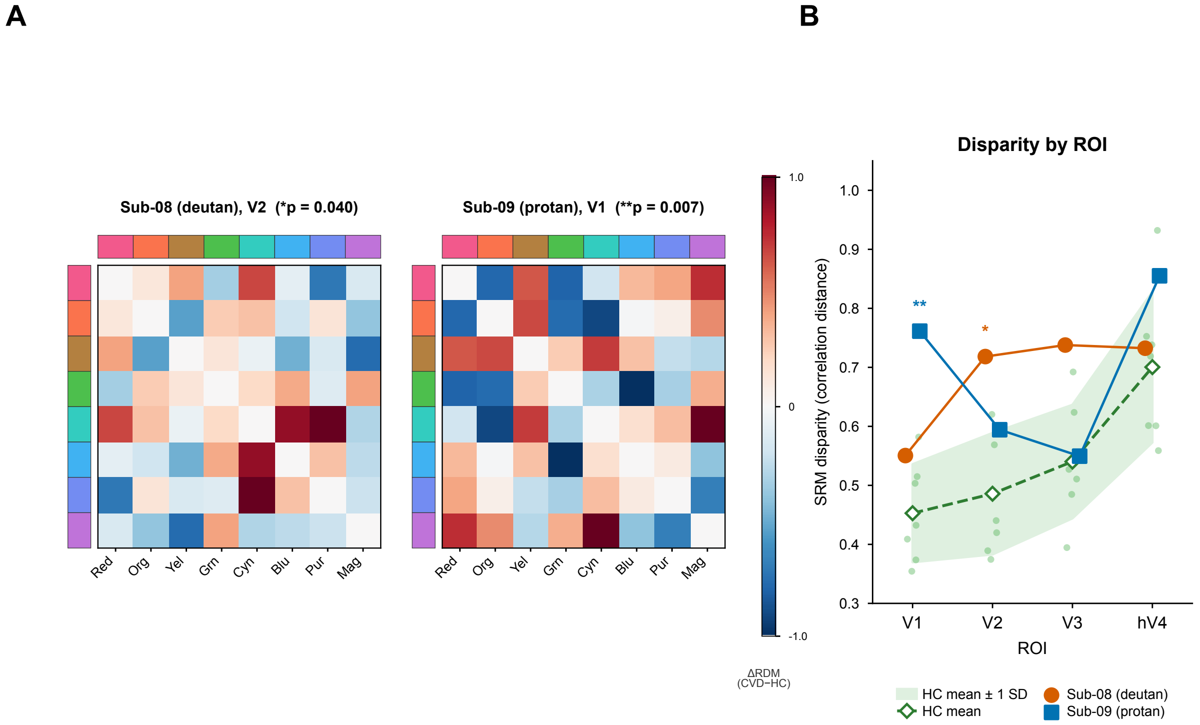

Figure 3. **Each CVD subject shows significantly elevated color representation disparity at a distinct ROI. (A)** $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC-LOO}}$ in SRM-aligned space at each subject's primary ROI (sub-08 V2; sub-09 V1). Warm: CVD distances larger than HC mean; cool: smaller. Annotated p-values: R+C cone-shift model permutation test against observed ΔRDM (sub-08 V2: $p = 0.179$, n.s.; sub-09 V1: $p = 0.026^{*}$). **(B)** Mean pairwise correlation distance (disparity) per subject per ROI. HC band: group mean ±1 SD ($n = 7$); individual HC LOO as dots. Crawford & Howell tests (sub-08 V2: $p = 0.040^{*}$; sub-09 V1: $p = 0.007^{**}$, exploratory single-case). Sub-10 (near-normal deutan, gray triangles): no significant elevation at any ROI. (A) and (B) are independent measures (model concordance vs. absolute disparity); significance patterns need not agree. Idiosyncratic ROI specificity (V2 deutan; V1 protan) is inconsistent with a shared group-level gain mechanism.

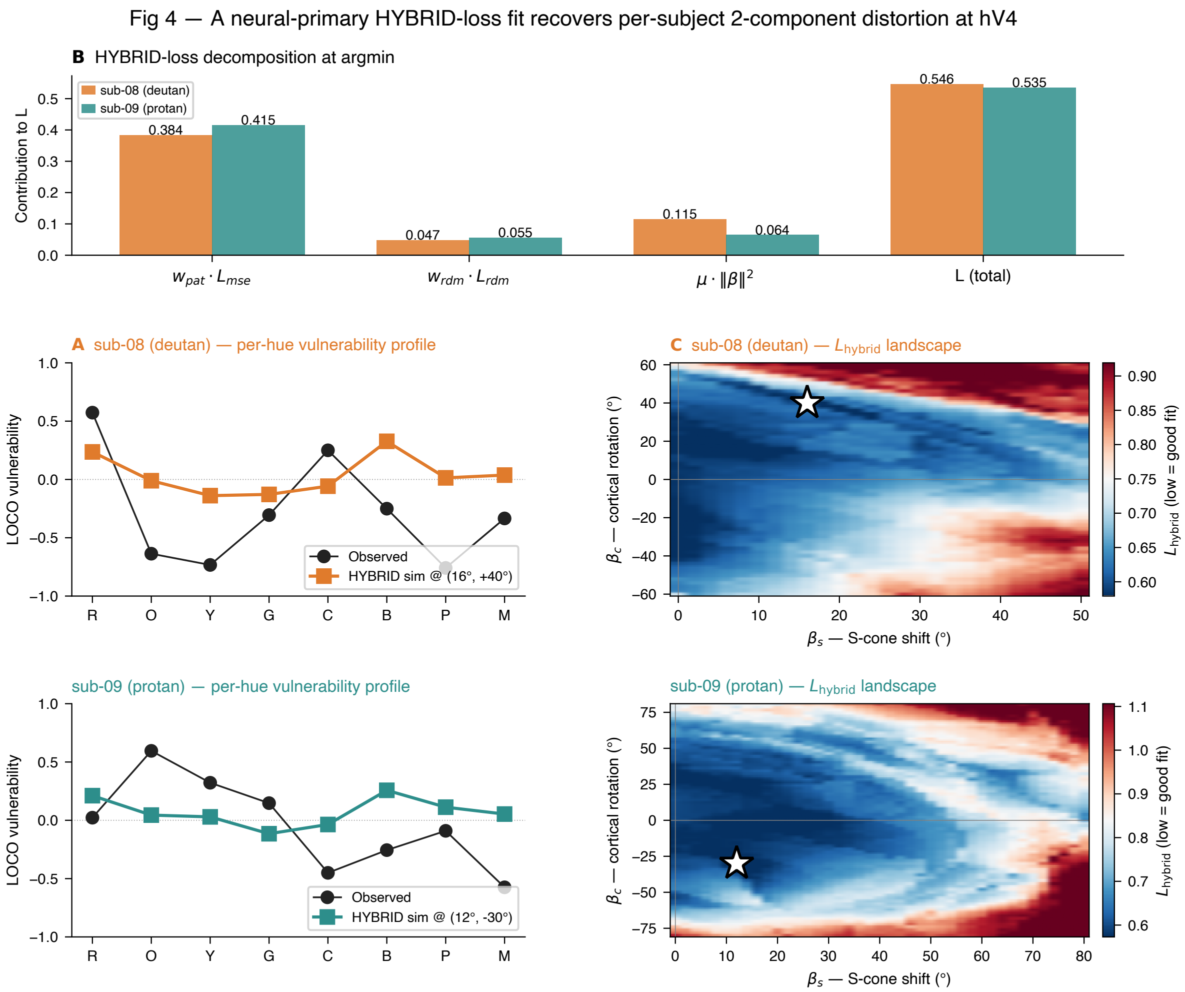

Figure 4. **A neural-primary HYBRID-loss fit recovers per-subject 2-component distortion at hV4.** Two rows: sub-08 (deutan), sub-09 (protan). **(A)** Per-hue hV4 LOCO vulnerability profiles. Filled circles: observed CVD vulnerability; coloured line: 2-component prediction at the HYBRID argmin. Hue labels (DKL, 45° spacing): R, O, Y, G, C, B, P, M. **(B)** HYBRID-loss component decomposition at the per-subject argmin (Equation 3). **(C)** HYBRID-loss landscape $L(\beta_s, \beta_c)$ over the $26 \times 51 = 1{,}326$ cell grid. White star: per-subject argmin (sub-08: $16°, +40°$; sub-09: $12°, -30°$). Caveats, consistency anchors, and the historical LOCO-ρ comparison are discussed in the main text (Section) and Appendix .

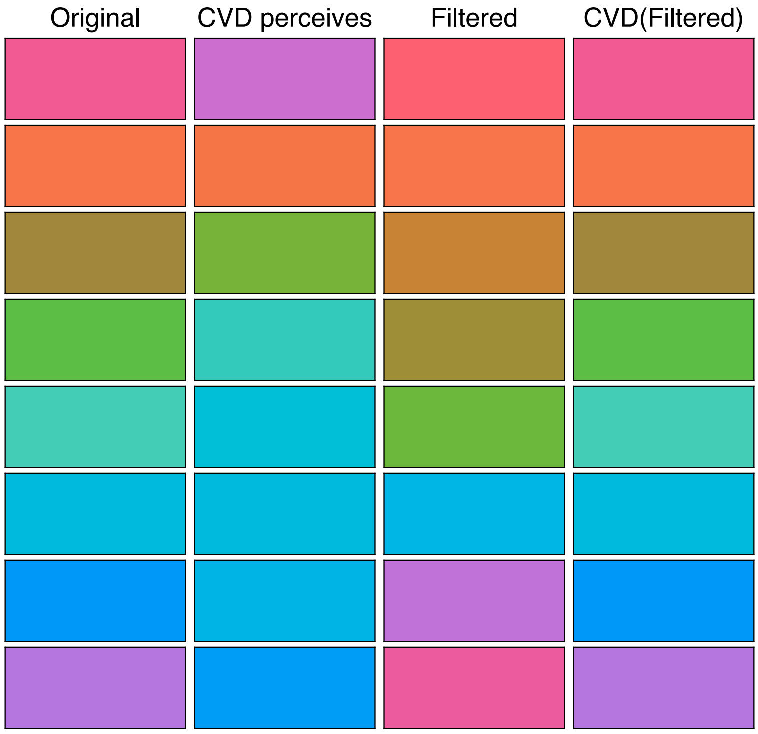

Figure 5. **HYBRID-derived per-subject stimulus-space filter: 4-column rendering, sub-08 (deutan).** Filter at $(\hat{\beta}_s, \hat{\beta}_c) = (16°, +40°)$, $\|\hat{\beta}\| = 43.1°$. Eight rows correspond to the eight displayed DKL hues. Columns (left to right): *Original* (HC percept of the displayed stimulus); *CVD perceives* (simulated CVD percept of the original under the per-subject 2-component model); *Filtered* (stimulus after applying $\delta\theta^{\text{filt}}$, the exact pre-image of $-\delta\theta(\hat{\beta}_s, \hat{\beta}_c)$, computed by Brent's-method refinement of the forward map; 8/8 exact, residual $< 0.001°$); *CVD(Filtered)* (simulated CVD percept of the filtered stimulus, qualitatively closer to the HC percept of the original).

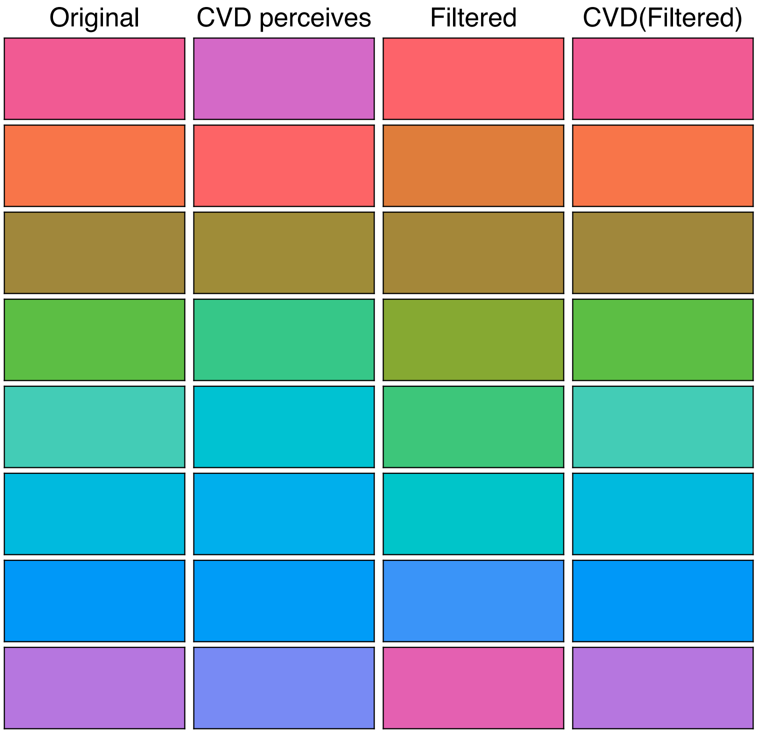

Figure 5. **(continued) Sub-09 (protan).** Filter at $(\hat{\beta}_s, \hat{\beta}_c) = (12°, -30°)$, $\|\hat{\beta}\| = 32.3°$. Columns and row order as in sub-08 above. Rendering coordinate space: STIM_LAB CIELab (project convention). Closeness of the right-most column to the left-most is *qualitative*; quantitative behavioural validation of filter efficacy is deferred to the Phase-3 2AFC arm (Methods , last paragraph). Comparison with the arc-compression failure of the 1-DOF Machado pre-image (sub-09 only) is reported in Appendix .